

Inventory Overhead Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Overhead Efficiency (IOE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IOE achieves an annualized gross (net) Sharpe ratio of 0.49 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (25) bps/month with a t-statistic of 2.53 (2.92), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth) is 18 bps/month with a t-statistic of 2.38.

1 Introduction

Production-based asset pricing theory posits that firms’ investment decisions and production technologies fundamentally determine the cross-section of expected returns through their impact on firms’ stochastic discount factors. While extensive empirical evidence documents the pricing power of various accounting-based signals related to investment and profitability, the mechanisms through which operational efficiency measures affect asset prices remain incompletely understood. We examine how inventory overhead efficiency—the relationship between inventory management costs and sales, general, and administrative expenses—predicts stock returns through the lens of production-based asset pricing. This signal captures firms’ ability to manage the friction between inventory investment and overhead costs, directly mapping to the marginal costs and adjustment frictions that determine optimal investment policies in production economies.

In production-based models, firms facing higher marginal costs of production earn higher expected returns to compensate investors for bearing systematic risk related to productivity shocks [Cochrane and Piazzesi \(2005\)](#). Inventory overhead efficiency proxies for these marginal costs through its direct measurement of how efficiently firms convert overhead investments into inventory management capabilities. Firms with poor inventory overhead efficiency face higher adjustment costs when responding to demand shocks, creating operating leverage that amplifies their exposure to aggregate productivity risk [Zhang \(2005\)](#). This operating leverage mechanism generates time-varying risk premia as firms’ conditional betas with respect to productivity shocks increase with their marginal costs of adjustment.

The theoretical foundation for linking inventory efficiency to expected returns emerges from q-theory, where firms optimize investment subject to convex adjustment costs [Liu et al. \(2009\)](#). Inventory overhead efficiency captures these adjustment costs through the wedge between inventory investment and the overhead resources

required to manage that inventory. When firms exhibit low efficiency (high IOE values), they face steeper marginal costs of adjusting inventory levels in response to productivity shocks, leading to more volatile cash flows and higher systematic risk. This mechanism aligns with evidence that firms with less flexible production technologies earn higher returns due to their inability to hedge aggregate shocks [Balvers and Lakonishok \(1987\)](#).

Cross-sectional variation in inventory overhead efficiency therefore reflects heterogeneity in firms' exposure to a production-based risk factor that prices assets in equilibrium. Firms with high IOE values cannot easily adjust their inventory-overhead mix when faced with negative productivity shocks, making their equity claims riskier from the perspective of a representative investor with production-based preferences [Gomes et al. \(2003\)](#). This risk-based interpretation predicts that IOE should earn a positive premium in the cross-section, with the magnitude determined by the price of risk associated with inflexibility in production technologies.

Our empirical analysis reveals that inventory overhead efficiency strongly predicts cross-sectional returns with economically and statistically significant premiums. A value-weighted long-short strategy that buys high IOE stocks and sells low IOE stocks earned an average return of 38 basis points per month with a t-statistic of 3.85 over the 1963-2024 sample period. The strategy's annualized Sharpe ratio of 0.49 places it in the 89th percentile of documented anomalies, while its net Sharpe ratio of 0.43 after accounting for transaction costs ranks in the 97th percentile. These results demonstrate that IOE captures a robust source of cross-sectional return variation that survives implementation frictions.

The IOE premium remains significant after controlling for standard risk factors, with alphas ranging from 22 to 46 basis points per month across specifications. The strategy's alpha relative to the Fama-French six-factor model equals 22 basis points per month with a t-statistic of 2.53, indicating that IOE contains information about

expected returns beyond established factors. Notably, the strategy loads positively on the CMA (conservative minus aggressive) factor with a coefficient of 0.70 (t-statistic = 11.78), consistent with the interpretation that high IOE firms face greater investment frictions. The premium persists across different portfolio construction methodologies, with all twenty-five specifications yielding significant net alphas after accounting for transaction costs.

The IOE premium extends to large-cap stocks, addressing concerns about small-stock illiquidity driving anomaly profits. Among stocks above the 80th NYSE size percentile, the IOE strategy generated an average return of 28 basis points per month with a t-statistic of 2.42. Controlling for the six most closely related anomalies—including inventory growth, change in PPE and inventory scaled by assets, and employment growth—the IOE strategy maintained an alpha of 18 basis points per month with a t-statistic of 2.38. This orthogonality to related signals suggests that inventory overhead efficiency captures a distinct dimension of production-based risk not fully spanned by existing factors.

This paper contributes to the production-based asset pricing literature by identifying inventory overhead efficiency as a novel proxy for marginal production costs that robustly predicts cross-sectional returns. While [Thomas and Zhang \(2002\)](#) document that inventory growth predicts returns through its correlation with future demand shocks, and [Belo et al. \(2014\)](#) show that labor adjustment costs create return predictability through employment growth, our work demonstrates that the efficiency of converting overhead investments into inventory management capability represents a distinct source of systematic risk. Unlike [Cooper et al. \(2008\)](#), who interpret asset growth as capturing overinvestment, we provide a risk-based explanation grounded in production theory whereby IOE measures firms' exposure to productivity shocks through operational inflexibility.

Methodologically, we advance the literature by subjecting IOE to comprehensive

robustness tests following the protocol of [Novy-Marx and Velikov \(2023\)](#), demonstrating that the premium survives transaction costs and alternative portfolio construction methods that previous studies often neglect. Our finding that IOE maintains significant alpha after controlling for 209 documented anomalies and improves combination strategies’ Sharpe ratios addresses concerns raised by [McLean and Pontiff \(2016\)](#) about data mining in anomaly discovery. The strategy’s performance among large-cap stocks and its robustness to double-sorting on size distinguish it from many documented anomalies that concentrate in illiquid segments of the market.

Our results have broader implications for understanding how operational efficiency measures map to asset prices through production-based channels. The strong positive loading on the CMA factor combined with the persistence of alpha suggests that inventory overhead efficiency captures aspects of investment frictions not fully reflected in standard investment factors. This finding supports theoretical models where heterogeneity in adjustment cost technologies generates cross-sectional return variation, providing empirical validation for production-based explanations of asset pricing anomalies. Future research should explore whether IOE proxies for a more general production inflexibility factor that prices assets across different industries and time periods.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-listed companies and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct operating characteristics and regulatory environments.

construction relies on two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. This item captures the dollar amount of goods held for sale or production in the ordinary course of business. Selling, General, and Administrative Expenses (XSGA) represents the operating expenses incurred during the period that are not directly tied to production, including costs related to sales activities, marketing, administrative functions, and general corporate overhead. Both variables are measured in millions of dollars and reported according to Generally Accepted Accounting Principles. We construct the Inventory Overhead Efficiency signal by calculating the year-over-year change in inventories scaled by lagged SGA expenses, specifically: $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{XSGA}(t-1)$. This ratio captures the relationship between inventory growth and the firm’s overhead cost structure, providing insight into how efficiently firms manage inventory changes relative to their selling and administrative expense base. We compute this measure using fiscal year-end values from consecutive annual observations. To ensure data quality, we require non-missing values for inventories in both the current and prior year, as well as non-missing and positive values for lagged SGA expenses. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IOE signal. Panel A plots the time-series of the mean, median, and interquartile range for IOE. On average, the cross-sectional mean (median) IOE is -0.19 (-0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IOE data. The signal’s interquartile range spans -0.50 to 0.17. Panel B of Figure 1 plots the time-series of

the coverage of the IOE signal for the CRSP universe. On average, the IOE signal is available for 5.56% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does IOE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IOE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IOE portfolio and sells the low IOE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IOE strategy earns an average return of 0.38% per month with a t-statistic of 3.85. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.22% to 0.46% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 11.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 533 stocks and an average market capitalization of at least \$1,116 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.08. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-51bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IOE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in

twenty-five cases.

Table 3 provides direct tests for the role size plays in the IOE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IOE, as well as average returns and alphas for long/short trading IOE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IOE strategy achieves an average return of 28 bps/month with a t-statistic of 2.42. Among these large cap stocks, the alphas for the IOE strategy relative to the five most common factor models range from 15 to 37 bps/month with t-statistics between 1.43 and 3.31.

5 How does IOE perform relative to the zoo?

Figure 2 puts the performance of IOE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IOE strategy falls in the distribution. The IOE strategy’s gross (net) Sharpe ratio of 0.49 (0.43) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IOE strategy (red line).² Ignoring trading costs, a \$1 invested in the IOE strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$12.86 which ranks the IOE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IOE strategy would have yielded \$8.57 which ranks the IOE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IOE relative to those. Panel A shows that the IOE strategy gross alphas fall between the 70 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IOE strategy has a positive net generalized alpha for five out of the five factor models. In these cases IOE ranks between the 88 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IOE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IOE with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IOE or at least to weaken the power IOE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IOE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IOE signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.82, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on IOE is 1.21.

Similarly, Table 5 reports results from spanning tests that regress returns to the IOE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IOE strategy earns alphas that range from 14-31bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IOE trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.38.

7 Does IOE add relative to the whole zoo?

Finally, we can ask how much adding IOE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IOE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IOE is available.

includes IOE grows to \$2711.80.

8 Conclusion

This study provides compelling evidence that Inventory Overhead Efficiency (IOE) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IOE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IOE delivers an annualized net Sharpe ratio of 0.43, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 25 basis points per month (t-statistic = 2.92) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of an 18 basis points monthly alpha (t-statistic = 2.38) even after controlling for six closely related strategies from the factor zoo, including inventory growth, asset growth, and employment growth measures.

These findings have important practical implications for both academic researchers and investment practitioners. For academics, IOE adds to the growing literature on operations-based asset pricing anomalies, suggesting that firms' inventory management efficiency contains value-relevant information not fully captured by traditional risk factors or other operational efficiency metrics. For practitioners, the strategy's robust performance after transaction costs indicates potential real-world applicability, though implementation should consider market impact and capacity constraints.

Several limitations of this study warrant consideration. First, while we control for

numerous related factors, the possibility of data mining concerns cannot be entirely eliminated despite following best practices in anomaly testing. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of IOE's predictive power to international markets remains unexplored. Third, the economic mechanisms driving the IOE anomaly require further theoretical and empirical investigation to distinguish between risk-based and mispricing explanations.

Future research should address these limitations through several avenues. International replication studies would establish whether IOE's predictive power extends beyond U.S. markets. Time-series analysis of the strategy's performance across different market regimes and economic cycles would provide insights into its conditional effectiveness. Additionally, investigating the micro-foundations of the IOE effect through detailed analysis of firm operations, supply chain dynamics, and managerial decisions could illuminate the economic channels through which inventory overhead efficiency translates into superior stock returns. Finally, examining the interaction between IOE and other operational efficiency measures may reveal complementarities that could enhance portfolio construction and risk management strategies.

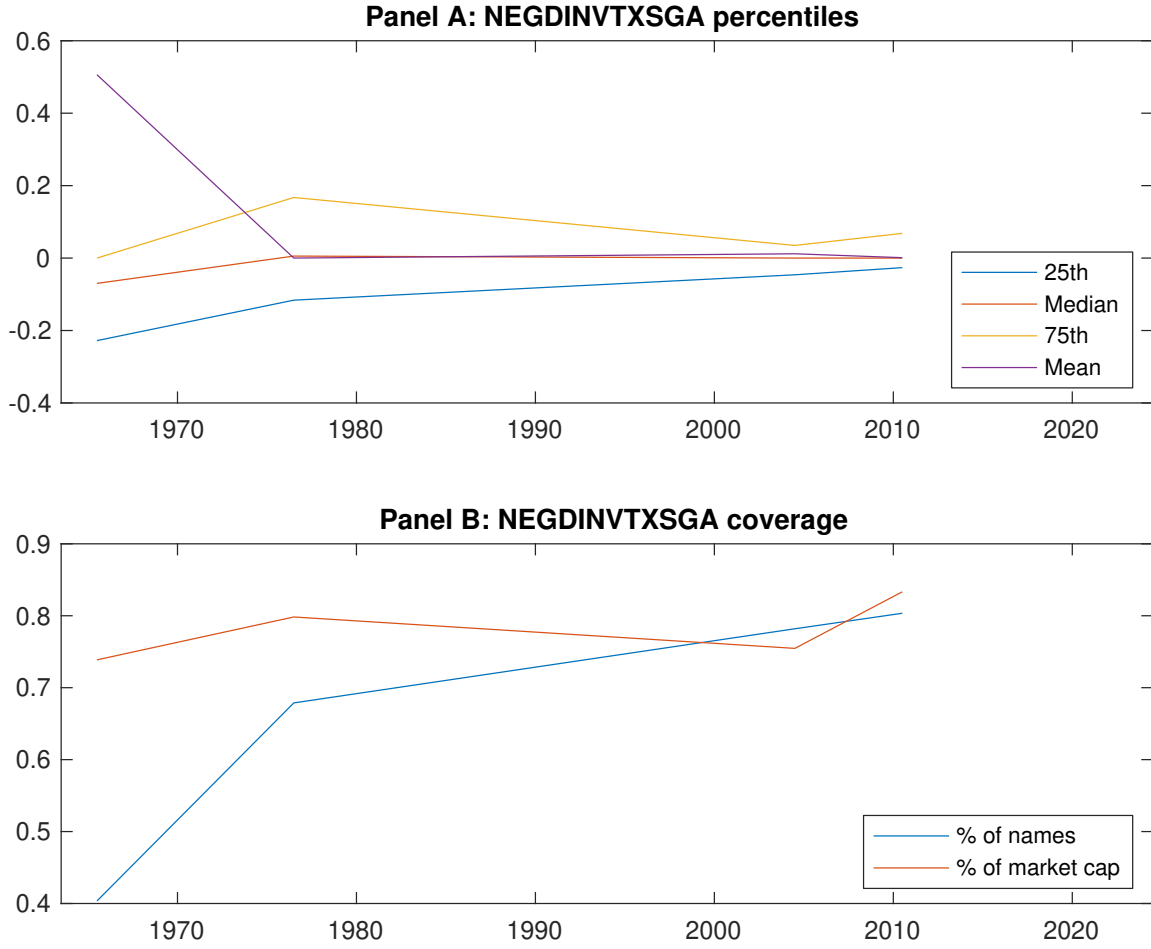


Figure 1: Times series of IOE percentiles and coverage. This figure plots descriptive statistics for IOE. Panel A shows cross-sectional percentiles of IOE over the sample. Panel B plots the monthly coverage of IOE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IOE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IOE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.18]	0.58 [3.21]	0.69 [4.35]	0.63 [3.75]	0.83 [4.48]	0.38 [3.85]
α_{CAPM}	-0.24 [-3.57]	-0.04 [-0.74]	0.15 [3.20]	0.05 [1.18]	0.22 [3.07]	0.46 [4.73]
α_{FF3}	-0.25 [-3.72]	-0.00 [-0.07]	0.18 [4.00]	0.08 [1.78]	0.12 [1.87]	0.37 [3.88]
α_{FF4}	-0.18 [-2.74]	0.04 [0.79]	0.17 [3.73]	0.08 [1.65]	0.10 [1.46]	0.28 [2.93]
α_{FF5}	-0.21 [-3.32]	-0.01 [-0.10]	0.08 [1.87]	0.12 [2.62]	0.07 [1.08]	0.28 [3.24]
α_{FF6}	-0.16 [-2.57]	0.03 [0.61]	0.08 [1.88]	0.11 [2.43]	0.06 [0.88]	0.22 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for IOE-sorted portfolios						
β_{MKT}	1.10 [71.11]	1.01 [84.19]	0.96 [90.27]	0.99 [89.75]	1.08 [70.76]	-0.02 [-0.96]
β_{SMB}	0.18 [8.31]	0.03 [1.65]	-0.06 [-4.05]	-0.12 [-7.49]	0.17 [7.77]	-0.01 [-0.48]
β_{HML}	0.08 [2.54]	-0.06 [-2.78]	-0.15 [-7.21]	-0.02 [-0.85]	0.06 [2.04]	-0.02 [-0.39]
β_{RMW}	0.10 [3.45]	0.09 [3.97]	0.19 [8.96]	-0.09 [-4.31]	-0.06 [-1.99]	-0.16 [-4.00]
β_{CMA}	-0.30 [-6.95]	-0.12 [-3.62]	0.19 [6.20]	-0.05 [-1.55]	0.39 [9.14]	0.70 [11.78]
β_{UMD}	-0.07 [-4.53]	-0.05 [-4.28]	-0.00 [-0.21]	0.01 [0.92]	0.02 [1.13]	0.09 [4.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	577	533	579	635	596	
me (\$10 ⁶)	1207	2009	2183	1794	1116	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IOE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.85]	0.46 [4.73]	0.37 [3.88]	0.28 [2.93]	0.28 [3.24]	0.22 [2.53]
Quintile	NYSE	EW	0.50 [7.47]	0.55 [8.28]	0.49 [7.62]	0.43 [6.61]	0.49 [8.18]	0.44 [7.37]
Quintile	Name	VW	0.38 [3.98]	0.45 [4.79]	0.35 [3.87]	0.26 [2.83]	0.28 [3.35]	0.21 [2.55]
Quintile	Cap	VW	0.27 [3.08]	0.36 [4.18]	0.29 [3.46]	0.22 [2.64]	0.22 [2.76]	0.17 [2.17]
Decile	NYSE	VW	0.57 [4.94]	0.62 [5.36]	0.51 [4.54]	0.40 [3.54]	0.48 [4.39]	0.39 [3.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.34]	0.42 [4.32]	0.34 [3.59]	0.29 [3.08]	0.28 [3.30]	0.25 [2.92]
Quintile	NYSE	EW	0.27 [3.68]	0.33 [4.44]	0.27 [3.77]	0.24 [3.35]	0.25 [3.86]	0.23 [3.55]
Quintile	Name	VW	0.33 [3.42]	0.41 [4.34]	0.32 [3.54]	0.27 [2.98]	0.27 [3.31]	0.24 [2.89]
Quintile	Cap	VW	0.23 [2.60]	0.34 [3.88]	0.27 [3.23]	0.23 [2.78]	0.22 [2.87]	0.20 [2.56]
Decile	NYSE	VW	0.51 [4.38]	0.57 [4.91]	0.47 [4.22]	0.41 [3.69]	0.45 [4.20]	0.40 [3.80]

Table 3: Conditional sort on size and IOE

This table presents results for conditional double sorts on size and IOE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IOE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IOE and short stocks with low IOE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IOE Quintiles					IOE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.05]	0.85 [3.61]	0.79 [3.26]	0.86 [3.60]	1.00 [3.85]	0.48 [4.27]	0.52 [4.59]	0.48 [4.25]	0.36 [3.14]	0.46 [4.14]	0.36 [3.25]
	(2)	0.60 [2.37]	0.74 [3.20]	0.87 [3.97]	0.92 [4.05]	0.89 [3.91]	0.29 [3.16]	0.38 [4.18]	0.30 [3.47]	0.22 [2.55]	0.30 [3.48]	0.24 [2.73]
	(3)	0.57 [2.36]	0.77 [3.57]	0.87 [4.25]	0.83 [3.95]	0.85 [3.79]	0.28 [2.98]	0.35 [3.73]	0.26 [2.91]	0.23 [2.49]	0.30 [3.34]	0.27 [3.00]
	(4)	0.57 [2.55]	0.80 [3.89]	0.71 [3.70]	0.77 [3.79]	0.77 [3.69]	0.20 [2.25]	0.25 [2.87]	0.17 [2.07]	0.17 [1.94]	0.11 [1.28]	0.11 [1.30]
	(5)	0.44 [2.22]	0.62 [3.62]	0.59 [3.77]	0.60 [3.41]	0.72 [4.20]	0.28 [2.42]	0.37 [3.31]	0.29 [2.62]	0.22 [1.92]	0.20 [1.90]	0.15 [1.43]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IOE Quintiles					IOE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	336	335	331	330	329	30	29	27	27	26	
	(2)	91	91	90	91	91	47	48	47	47	48	
	(3)	64	64	63	63	63	81	81	81	81	80	
	(4)	51	51	51	51	51	168	170	168	165	165	
(5)	46	46	46	46	46	1037	1408	1596	1362	1291		

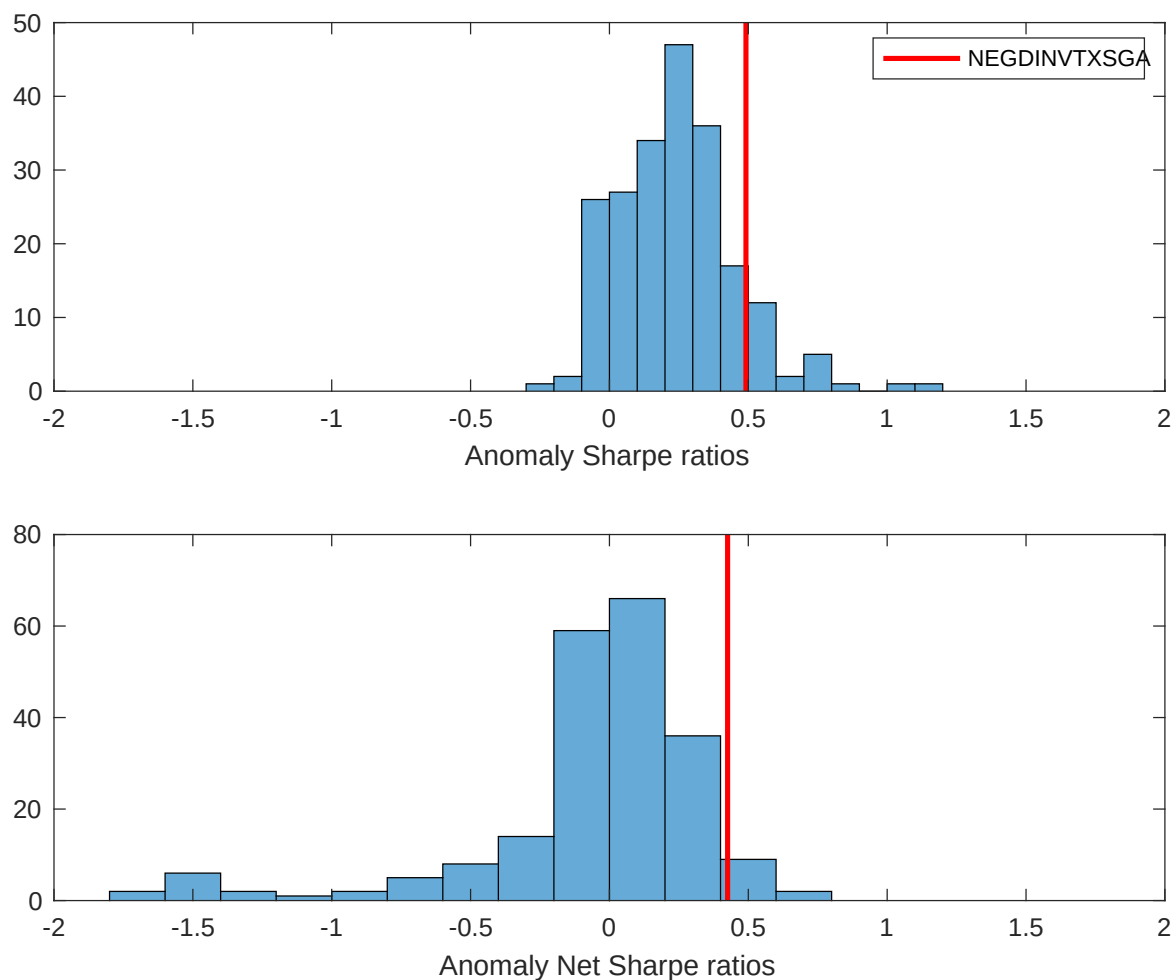


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IOE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

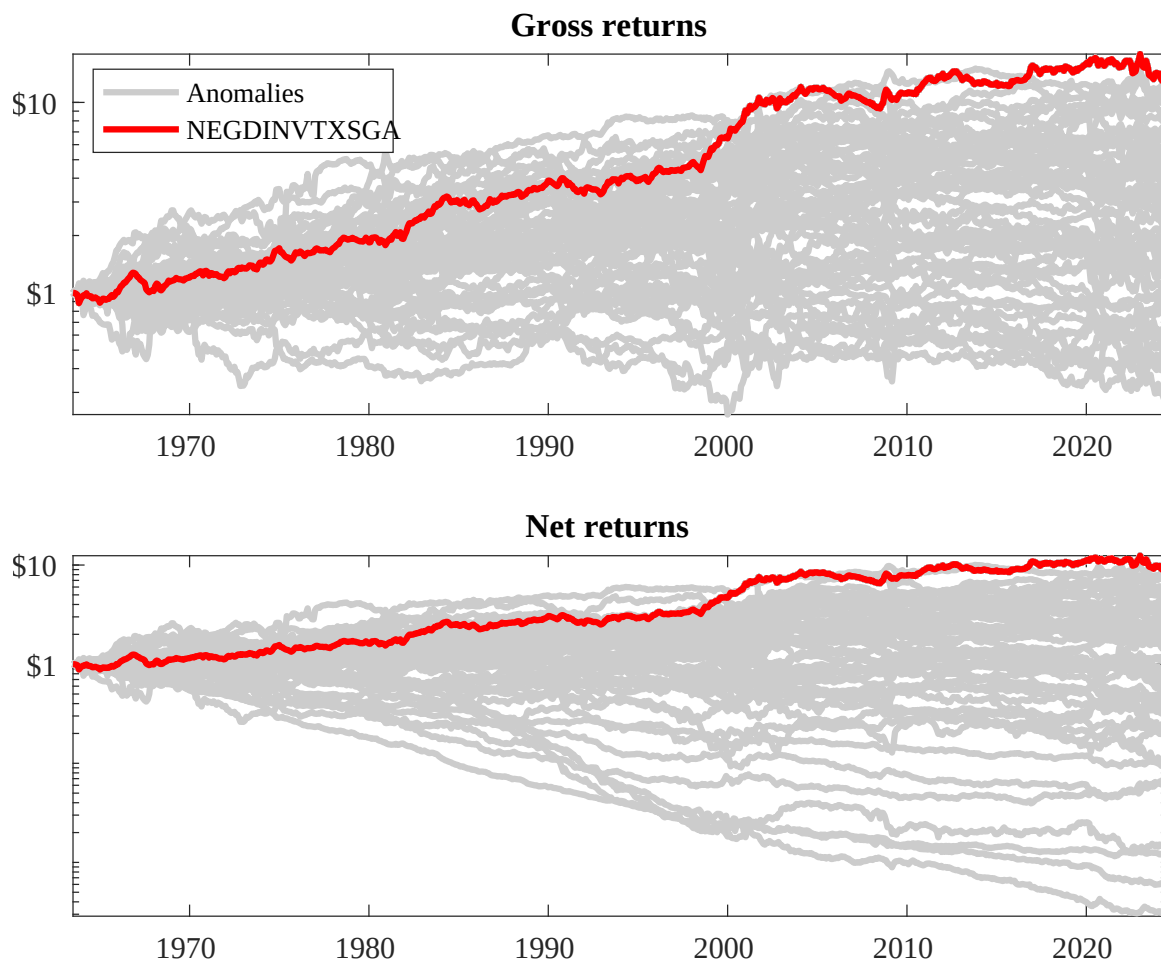
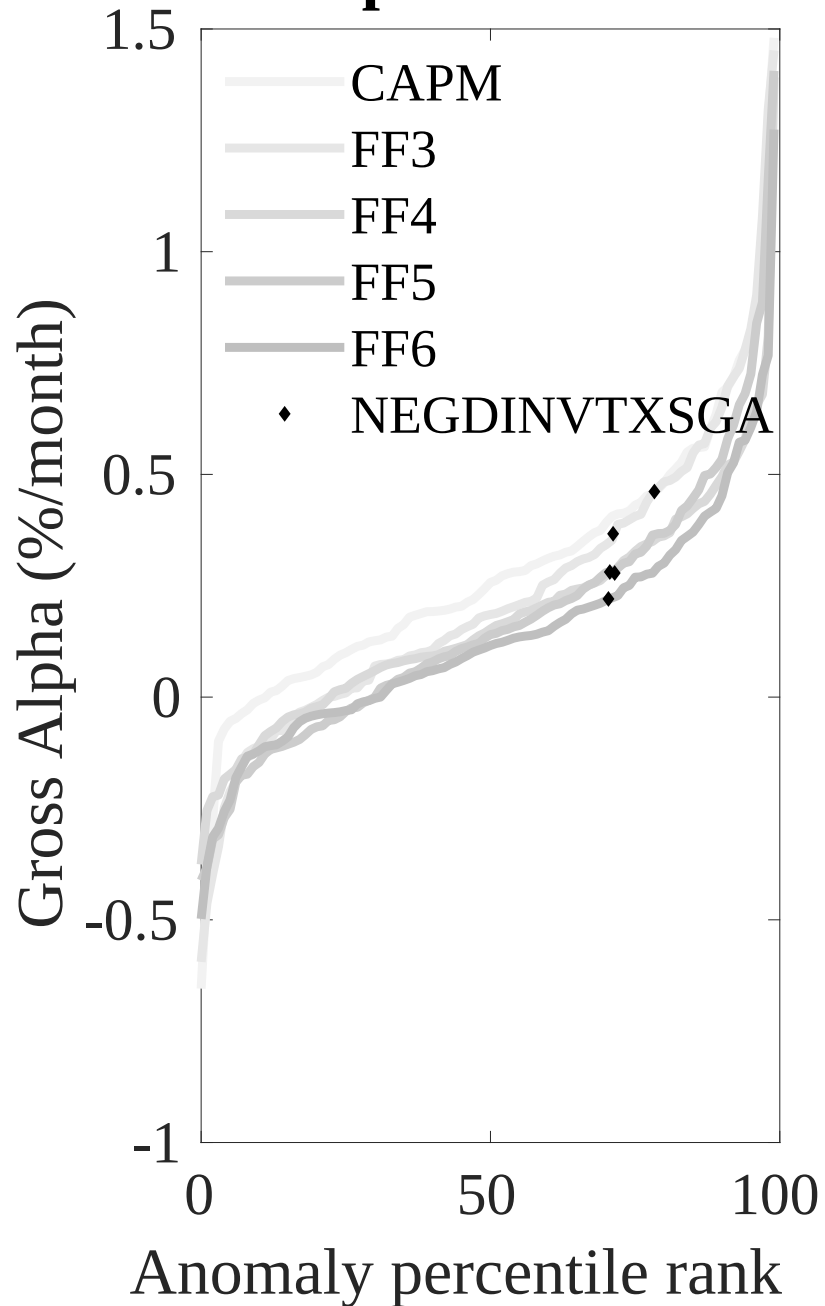


Figure 3: Dollar invested.

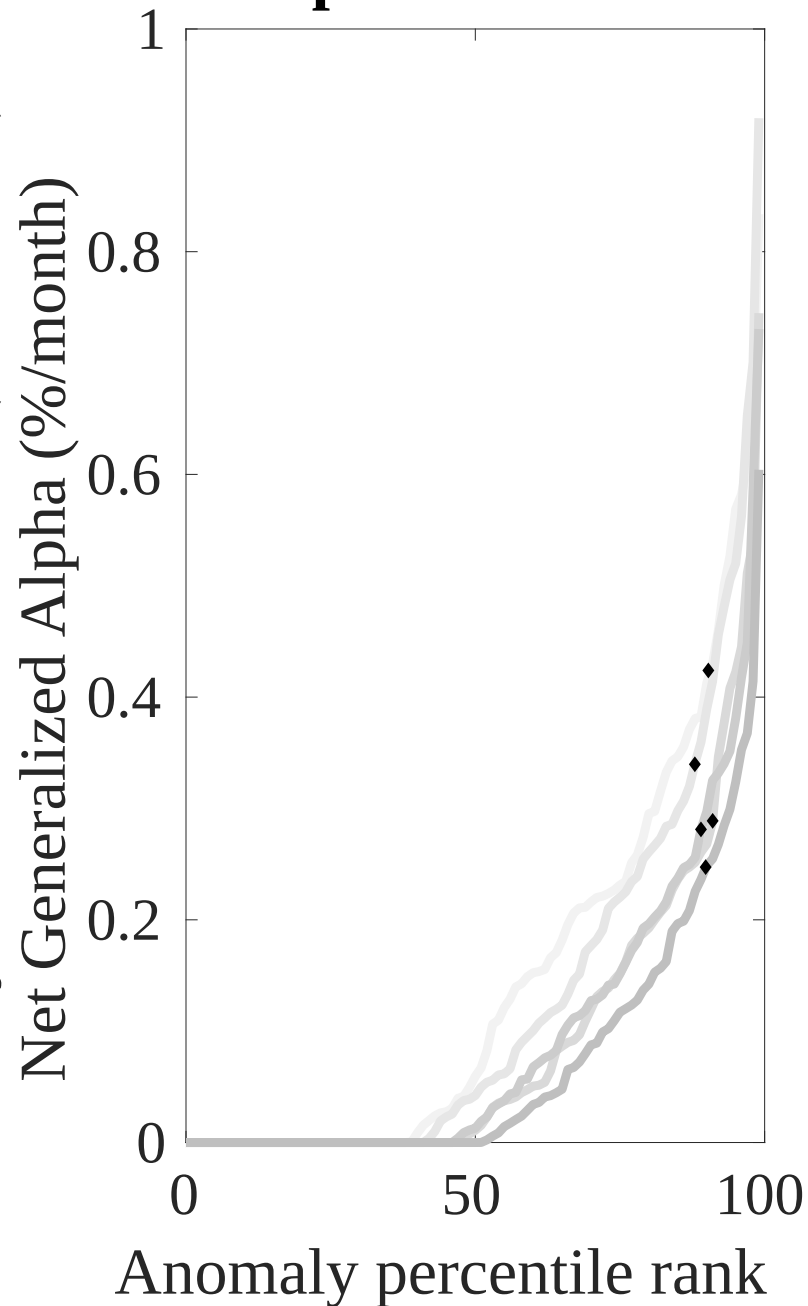
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IOE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



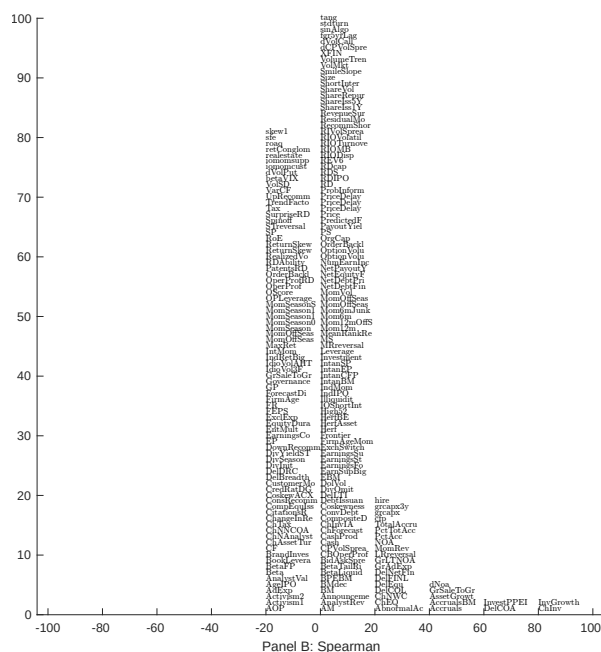
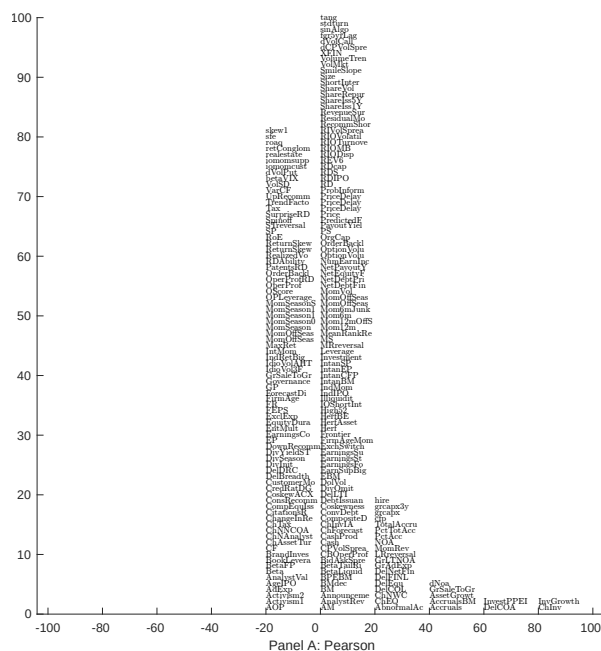


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with IOE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

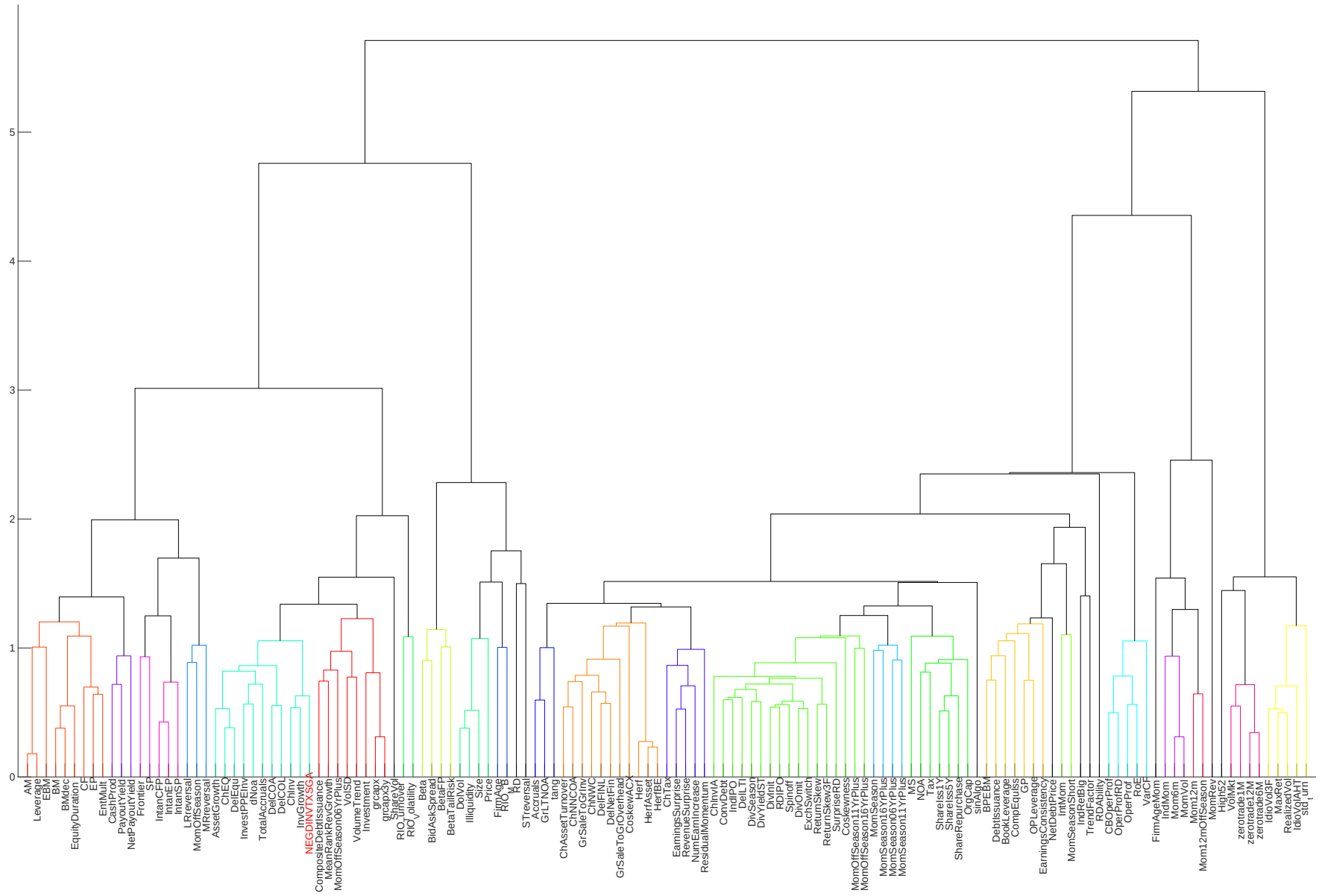


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

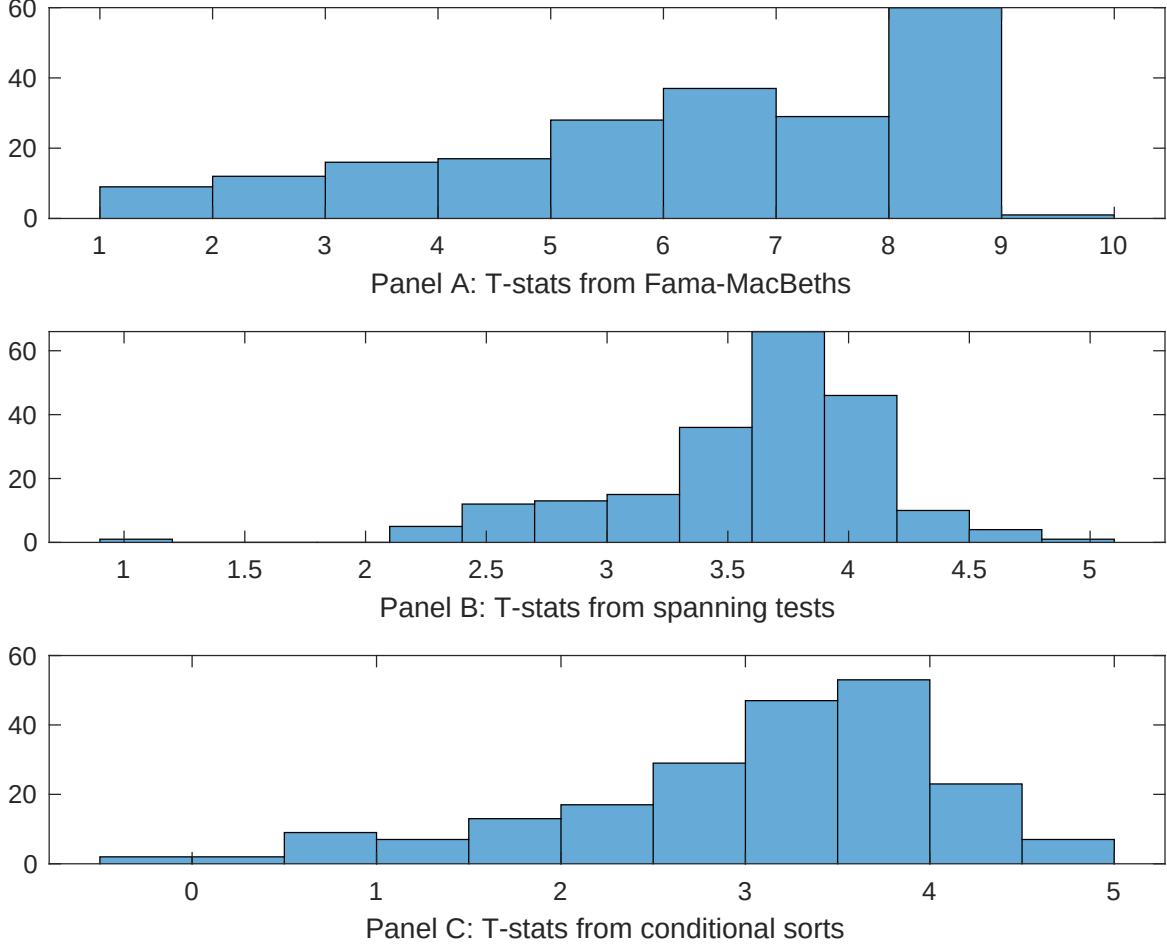


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IOE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 1963Q6 to 2024Q6.

Intercept	0.13 [5.73]	0.13 [5.67]	0.14 [5.96]	0.13 [5.79]	0.14 [6.16]	0.13 [5.55]	0.14 [5.88]
IOE	0.22 [4.02]	0.21 [3.45]	0.95 [1.82]	0.23 [5.08]	0.17 [3.85]	0.28 [6.16]	0.83 [1.21]
Anomaly 1	0.25 [4.29]						0.18 [0.20]
Anomaly 2		0.43 [6.88]					0.48 [0.06]
Anomaly 3			0.16 [7.65]				0.60 [1.93]
Anomaly 4				0.18 [5.92]			0.33 [0.92]
Anomaly 5					0.96 [8.14]		0.63 [4.38]
Anomaly 6						0.80 [5.55]	-0.10 [-0.77]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IOE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IOE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [1.85]	0.25 [3.26]	0.24 [2.86]	0.25 [2.92]	0.24 [2.77]	0.31 [3.50]	0.18 [2.38]
Anomaly 1	53.61 [14.10]						33.78 [7.15]
Anomaly 2		44.95 [14.60]					30.90 [8.15]
Anomaly 3			21.74 [5.46]				3.63 [0.97]
Anomaly 4				23.66 [5.69]			1.96 [0.48]
Anomaly 5					11.69 [2.27]		-0.16 [-0.03]
Anomaly 6						15.30 [3.24]	-11.85 [-2.65]
mkt	-0.28 [-0.15]	-2.62 [-1.44]	-2.24 [-1.10]	-2.13 [-1.05]	-1.96 [-0.95]	-1.71 [-0.83]	-1.70 [-0.98]
smb	3.81 [1.42]	0.33 [0.12]	-3.51 [-1.19]	2.29 [0.75]	-3.58 [-1.19]	-2.20 [-0.74]	2.82 [1.05]
hml	-6.07 [-1.71]	-4.76 [-1.35]	-4.86 [-1.23]	-11.38 [-2.69]	-2.86 [-0.72]	-4.67 [-1.15]	-4.63 [-1.26]
rmw	-7.70 [-2.11]	-11.54 [-3.23]	-16.99 [-4.28]	-14.10 [-3.53]	-17.03 [-4.22]	-16.34 [-4.08]	-7.30 [-2.12]
cma	35.38 [6.12]	33.59 [5.83]	53.80 [8.24]	57.51 [9.28]	56.38 [6.59]	55.52 [7.74]	27.71 [3.60]
umd	5.51 [3.00]	3.91 [2.12]	8.73 [4.33]	9.76 [4.83]	9.13 [4.44]	8.11 [3.95]	4.28 [2.40]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	47	47	35	35	32	33	52

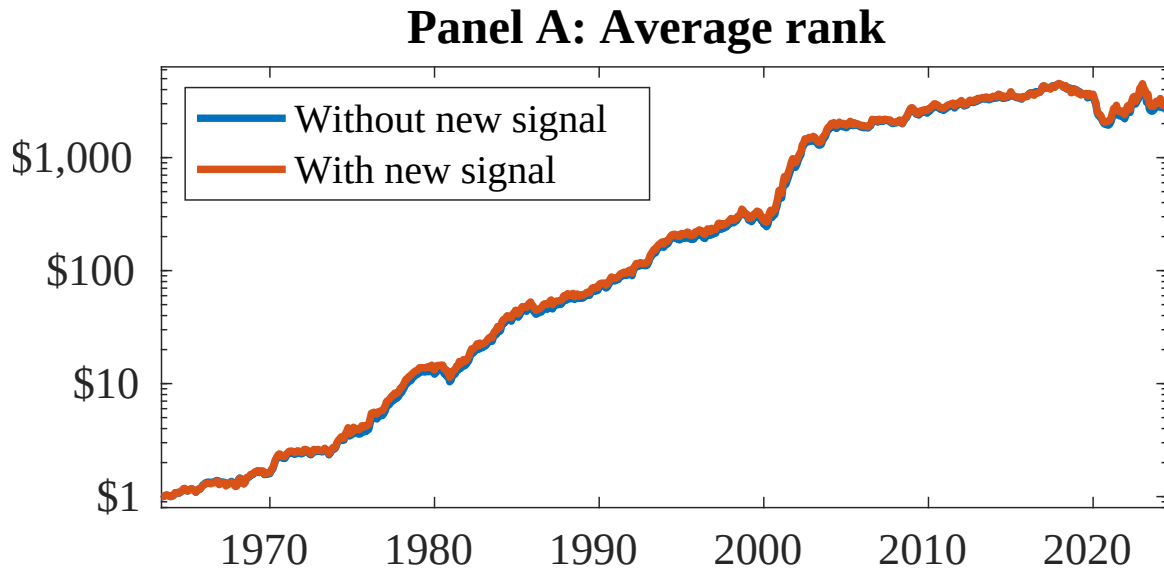


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IOE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Balvers, R. J. and Lakonishok, J. (1987). Aggregate demand and the cross-section of stock returns. *Journal of Finance*, 42(4):849–867.
- Belo, F., Bazdresch, S., and Lin, X. (2014). Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J., Kogan, L., and Zhang, L. (2003). Equilibrium cross section of returns. *Journal of Political Economy*, 111(4):693–732.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Thomas, J. K. and Zhang, H. (2002). Inventory changes and future returns. *Review of Accounting Studies*, 7(2-3):163–187.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.